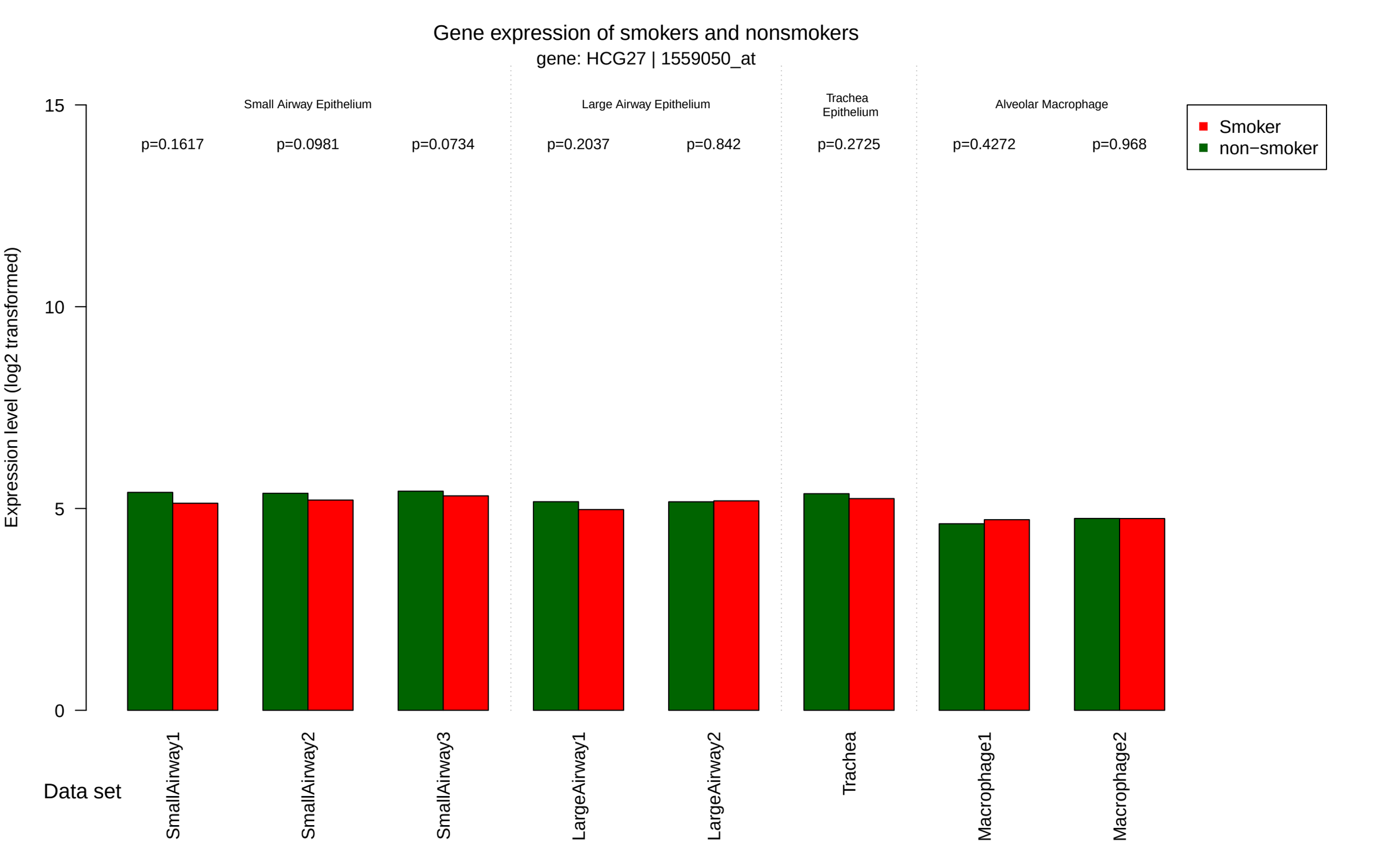
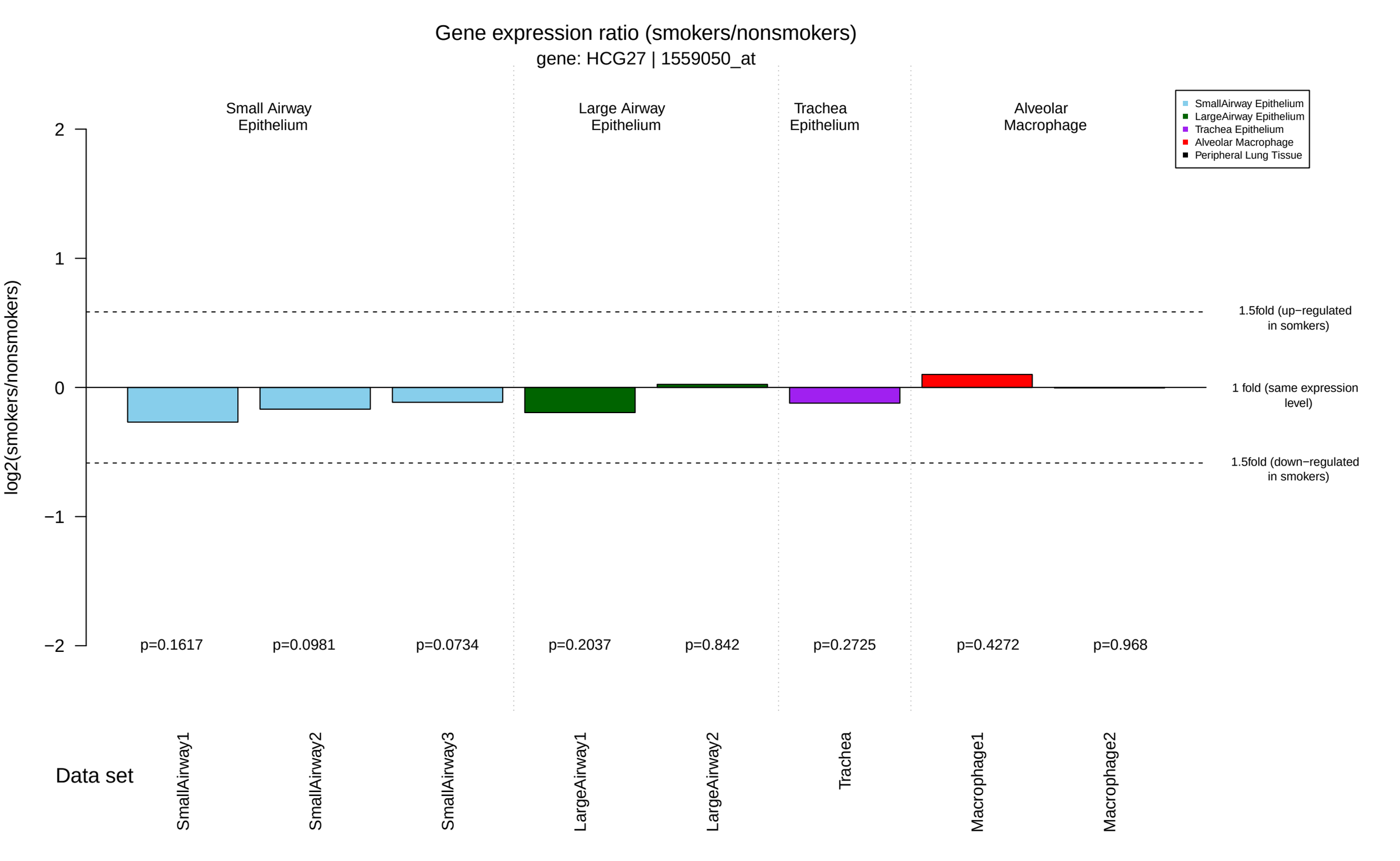
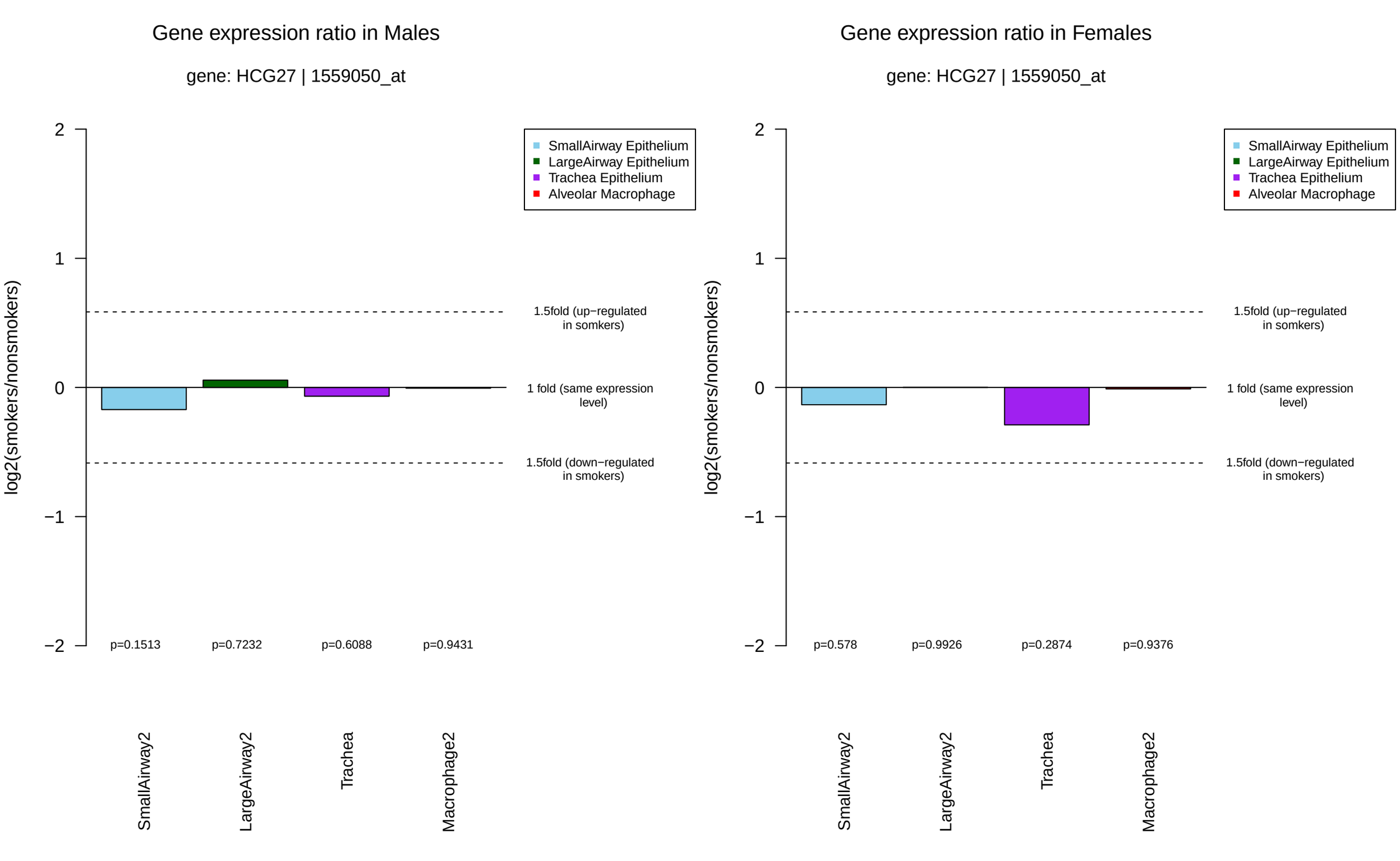
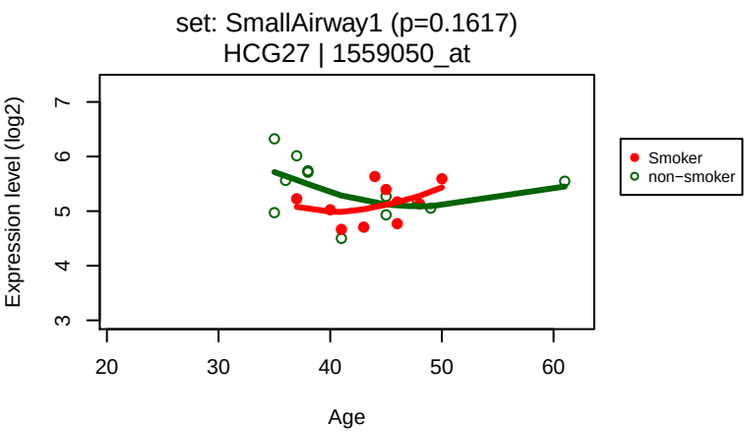
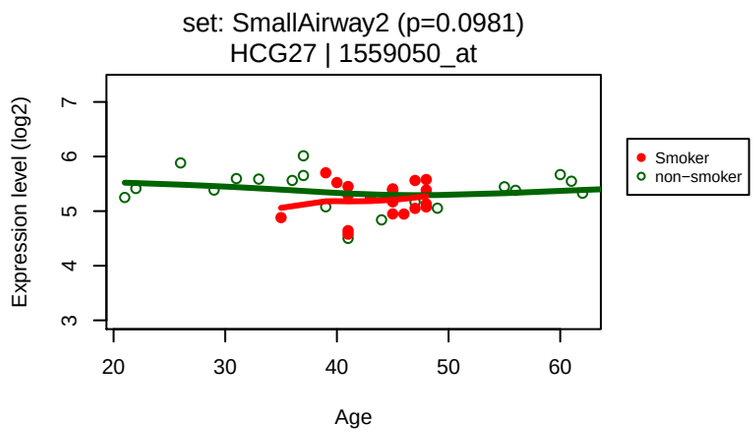
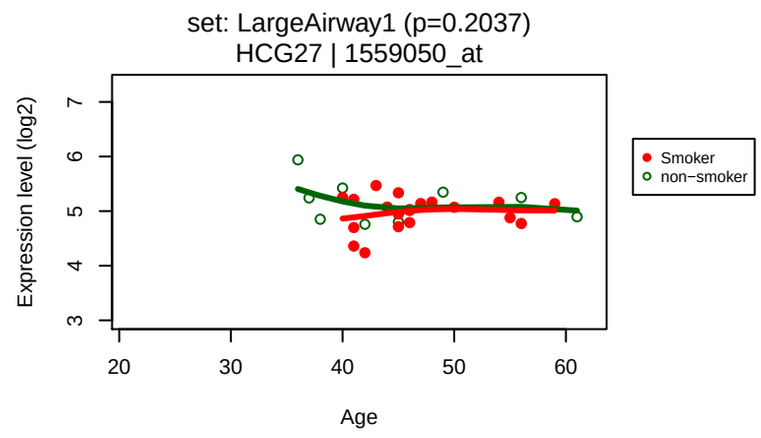

Gene expression of smokers and nonsmokers
gene: HCG27 | 1559050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1617
p=0.0981
p=0.0734
p=0.2037
p=0.842
p=0.2725
p=0.4272
p=0.968
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

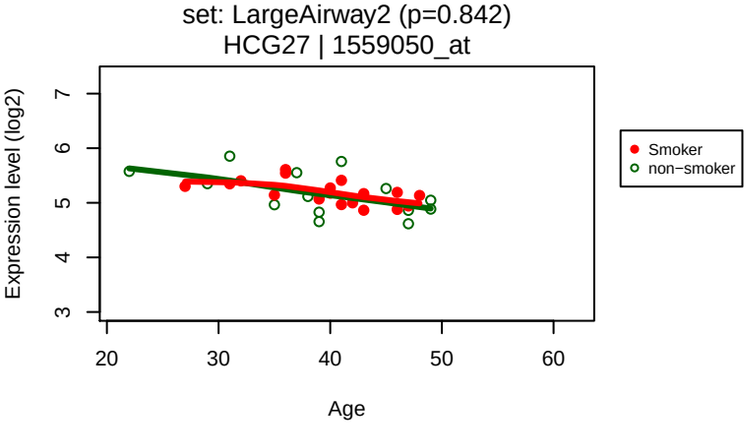

Gene expression ratio (smokers/nonsmokers)
gene: HCG27 | 1559050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1617
p=0.0981
p=0.0734
p=0.2037
p=0.842
p=0.2725
p=0.4272
p=0.968
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

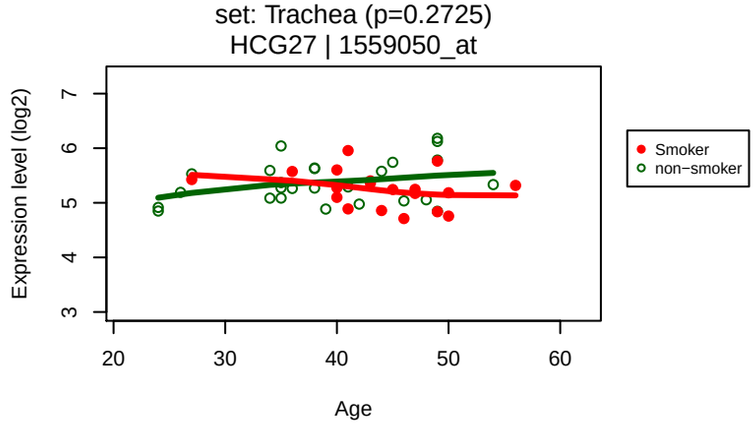

Gene expression ratio in Males
gene: HCG27 | 1559050_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1513
p=0.7232
p=0.6088
p=0.9431
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCG27 | 1559050_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.578
p=0.9926
p=0.2874
p=0.9376
SmallAirway2
LargeAirway2
Trachea
Macrophage2

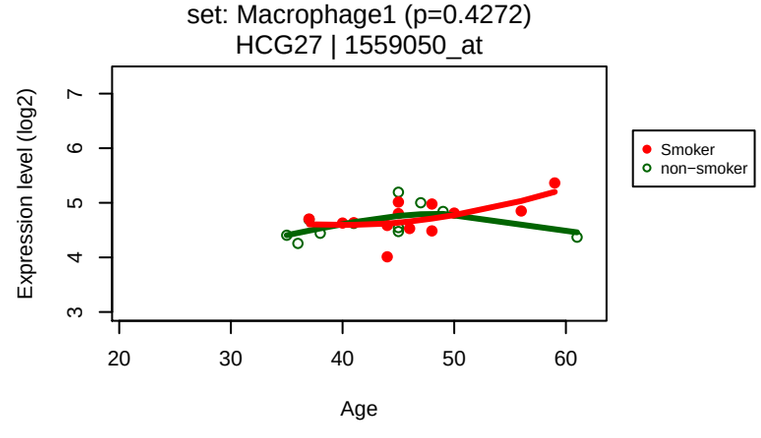

set: SmallAirway1 (p=0.1617)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker

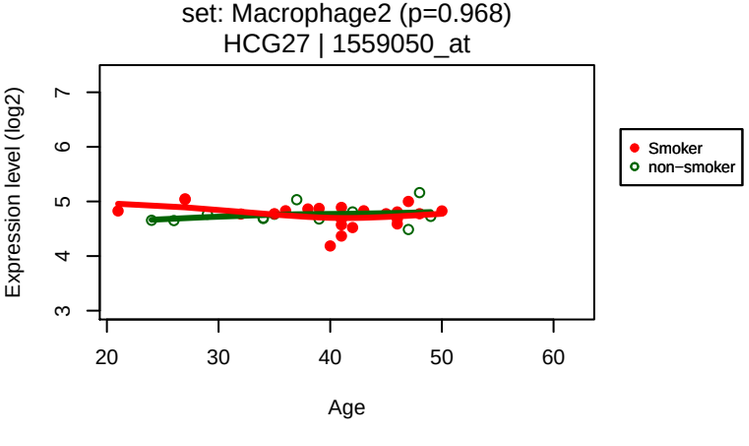

set: SmallAirway2 (p=0.0981)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker

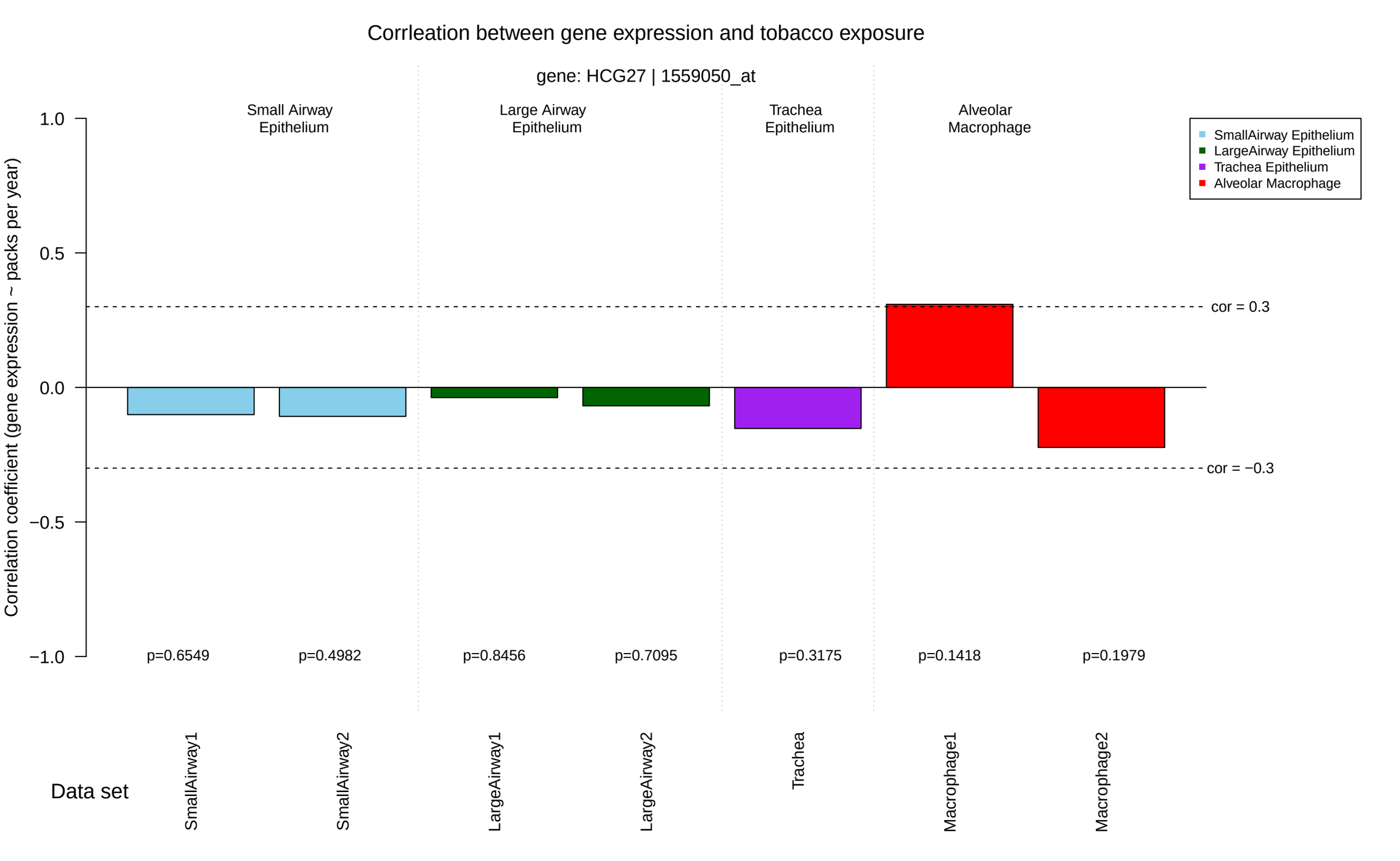

set: LargeAirway1 (p=0.2037)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.842)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2725)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4272)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.968)
HCG27 | 1559050_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCG27 | 1559050_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6549
p=0.4982
p=0.8456
p=0.7095
p=0.3175
p=0.1418
p=0.1979
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2